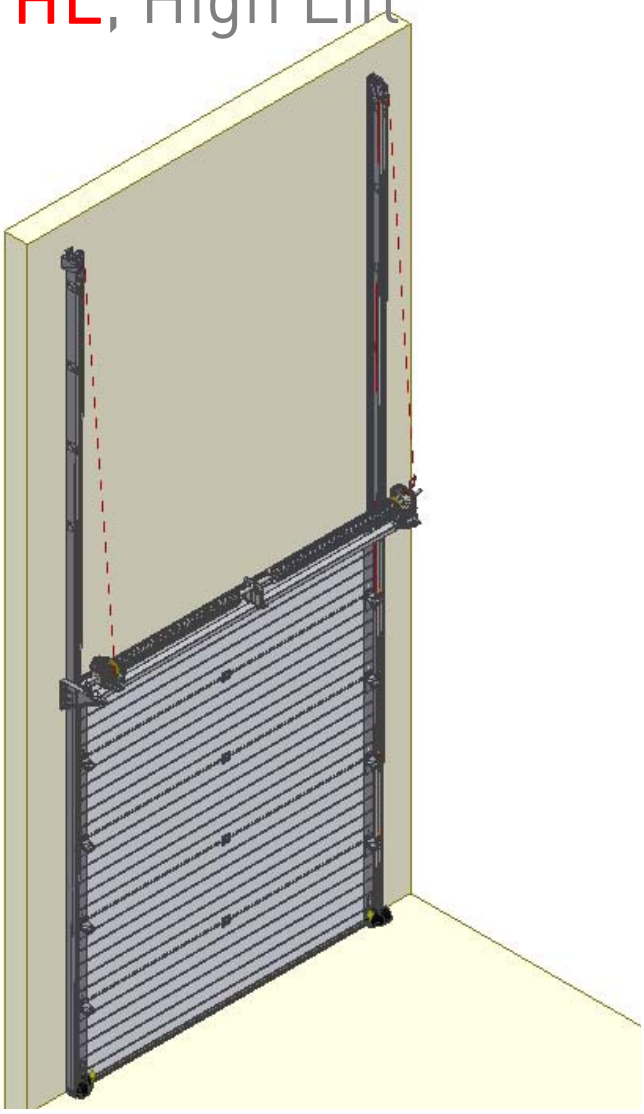
DockDoor

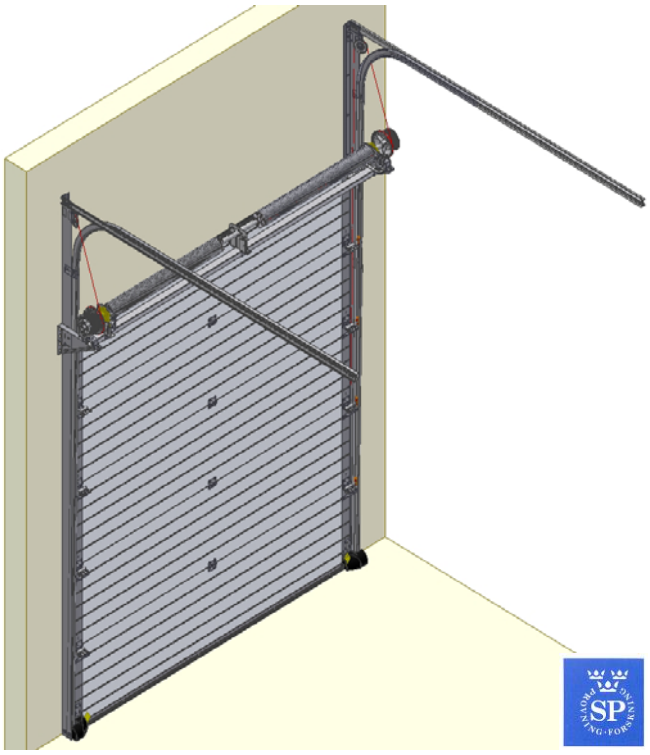
CE

Hardware system for overhead doors

VL, Vertical Lift

HL, High Lift





ATTENTION! GENERAL WARNING!

To install, use and maintain this door system safely, a number of precautions must be taken. For the safety of all concerned, pay heed to the warnings and instructions given below! If in doubt, contact your supplier.

SPECIAL SAFETY WARNINGS OR REMARKS IN THIS MANUAL ARE INDICATED WITH THIS SYMBOL. READ THESE WARNINGS CAREFULLY!

- ✓ This manual has been written for use by experienced fitters and as such is not suitable for d.i.y. purposes or for use by trainee fitters.
- ✓ This manual describes the installation of the hardware set components and door sections (panels). Be sure to supplement this manual if needed with instructions for any additional components not described in this manual.
- ✓ Before starting, read this manual carefully!
- ✓ Certain components may be sharp or have jagged edges. As such you are advised to wear safety gloves.
- ✓ All the components which have been supplied are designed for use with this specific overhead door. Replacement or adding additional components may have an adverse effect on the safety of, and the guarantee on, the door. Also the CE-approval which has been granted to this door will be cancelled when components are changed or installation is not done according to this manual! Installer is responsible for this.
- ✓ Ensure that there is sufficient light during installation. Remove obstacles and dirt. Make sure that there is no one else present other than the fitters. Other people (children!) may get in the way or endanger themselves during the installation.

GUARANTEE, CONDITIONS AND TERMS

The general terms and conditions of delivery and payment issued by the Metaalunie and designated as METAALUNIE CONDITIONS are fully applicable to all our quotations, contracts and their implementation. We expressly reject all other terms and conditions. On request we will send you a copy of these terms and conditions free of charge. A copy may also be downloaded from our website www.flexiforce.com.

FlexiForce strives to deliver 100 % in conformance with the order. In practice, in spite of all our controls, this is not always possible. However we will rectify any errors as quickly as possible, in order to minimise the inconvenience caused to you or the user. As such, it is important that you inform us as soon as possible about any problem with the delivery (include the order number and week of production) and give us the opportunity to offer a suitable solution.
FlexiForce will only reimburse third party costs if we have given explicit permission for this in advance. The reimbursement is based on normal rates and travelling expenses over distances of 1 hour away at most.
For large-scale projects we strongly advise you to first install 1 door completely before installing the other doors. In this way, any errors can be detected early on and rectified comparatively cheaply.
This manual does not confer any rights. Technical modifications may be made without written notice.
FlexiForce has endeavoured to design and put together this hardware set in conformance with the applicable CE-norms. However, we recommend to check our configuration against any local national specification.

FlexiForce has applied the mandated INITIAL TYPE TESTING for doors described in this manual, at the SP-Institute in Sweden (as Notified Body Nr. 0402). The INITIAL TYPE TESTING REPORT that has been rewarded, can be transferred to the door producing company. This is needed to complete your CE files according to product standard EN13241-1. ITT REPORT 0402-CPD-402201.

INTRODUCTION

Flexi-Force has developed a hardware set for DockDoors. The set includes the following features:

- **Suitable for days opening W x H = 3250 x 3300mm**
- **Advised door weight max. 165 kg**
- **CE approved when selecting the proper Flexi-Force components**

In this handbook we shall restrict our instructions for the proper assembly of the complete door. For installation of not mentioned products or parts, including any components added by the supplier, as well as for a user handbook, we refer you to the supplier of the complete overhead door who is also responsible for the correct CE marking of the door. These instructions have been prepared for use by experienced professional fitters and are therefore not suitable for the "DIY" fitter or the apprentice fitter.
The article codes of the parts are given in (parentheses). This door can only be installed according to the CE-certificate if all parts are according to the parts lists of the SP-certificate.
See www.flexiforce.com.

NB! This garage door cannot be equipped with a pass door (wicket door). All CE-approvals are not valid when a pass door is installed. If you want to include a pass door, a new Initial Type Testing should be arranged under your responsibility.

RECOMMENDED ADDITIONAL MANUALS





IN THIS MANUAL:

- Attention! General Warning! 2
- Guarantee, conditions and terms 2
- Introduction 3
- Recommended additional manuals 3
- Tools for correct and rapid assembly 5
- Applied fixation materials 5
- System overview 6
- Installation Vertical Lift rail system 8
- Installation Horizontal Lift rail system 10
- Panel <> Hardware installation 11
- Installation shaft system 13
- Installation spring bumper High Lift 13
- Cables and torsion springs 14
- Finishing the door 17
- Panel assembly 19
- Trouble shooting 22
- Dismanteling the overhead door 23
- Maintenance and replacement of parts 24



TOOLS FOR CORRECT AND RAPID ASSEMBLY

(Battery) drill with	Bit 4,0 mm
	Plug 10 mm
	Plug 13 mm
Hexagonal key	4 mm
Socket screw spanner	8mm (in)
Ring /open ended spanner	10 mm
Ring /open ended spanner	13 mm
Ring /open ended spanner	15 mm
Ring /open ended spanner	17 mm
Socket wrench	with ¼" square
Wrench	
Gluing clamp	
Cord	
Water level (hose)	
2 blocks of ca. 20 en 40mm in height	
Measuring equipment	

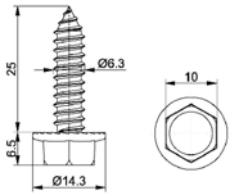
APPLIED FIXATION MATERIALS

The applicable fixation material is described in the drawings of this manuals. Below an overview of the applied fasteners and optional alternatives.

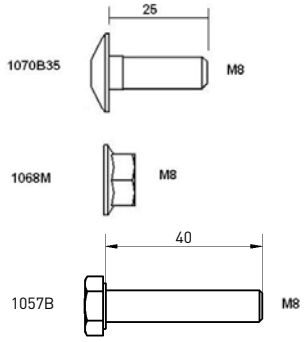
1055BV, self-tapping screw
6,3x16mm
Max torque 10Nm, key 10
6,3x35mm

Alternatives: 1052BV, screw

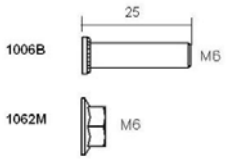
1053BV, screw



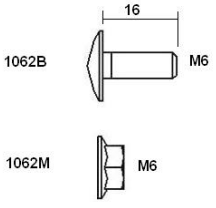
1070B35, bolt M8x25
1068M, nut M8
Max. torque 13Nm, key 13



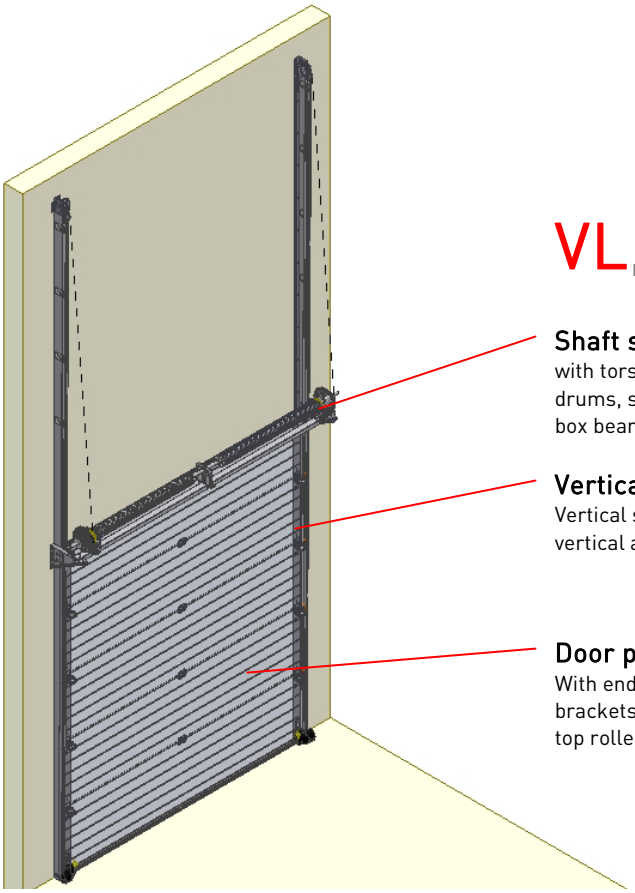
1006B, press bolt
1062M, nut M6
Max. torque 9,1Nm, key 10



1062B, bolt M6x16
1062M, nut M6
Max. torque 5,3Nm, key 10



SYSTEM OVERVIEW

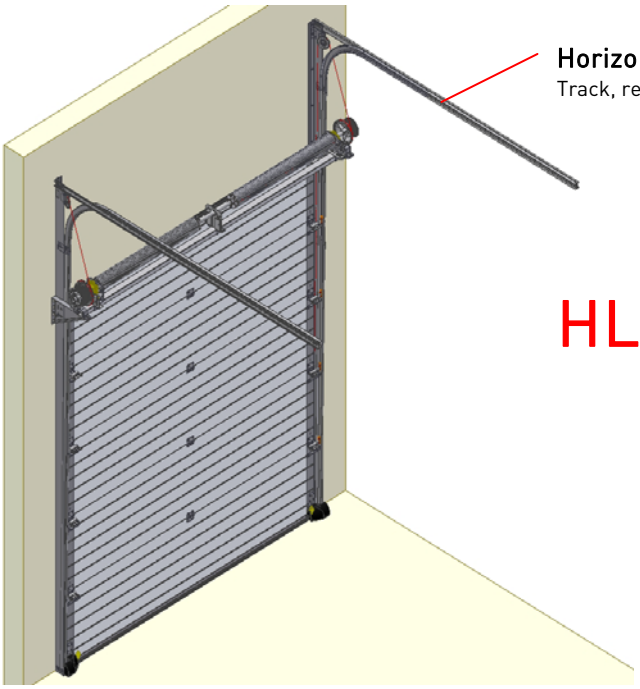


VL, Vertical Lift

Shaft system
with torsion springs, spring break devices drums, spring fittings, etc pre-assembled on box beam

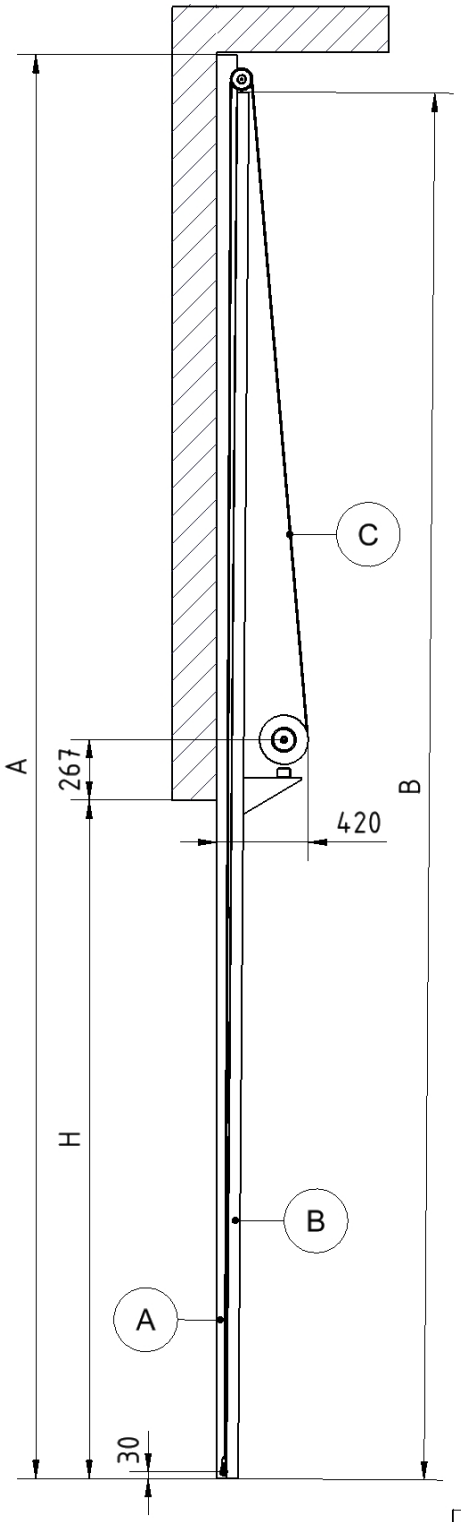
Vertical track
Vertical set with track, side seal and vertical angle; spot welded

Door panels, sections
With end caps, hinges, bottom brackets, bottom astragal and seal, top roller carriers, windows, etc.



Horizontal track system
Track, reinforcement and curve.

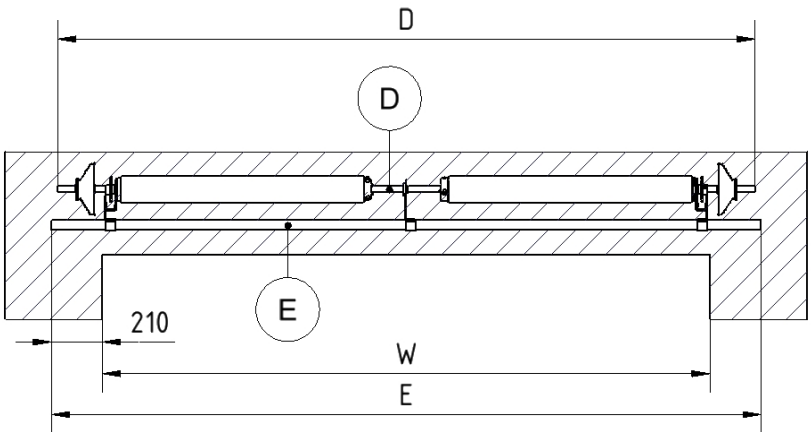
HL, High Lift

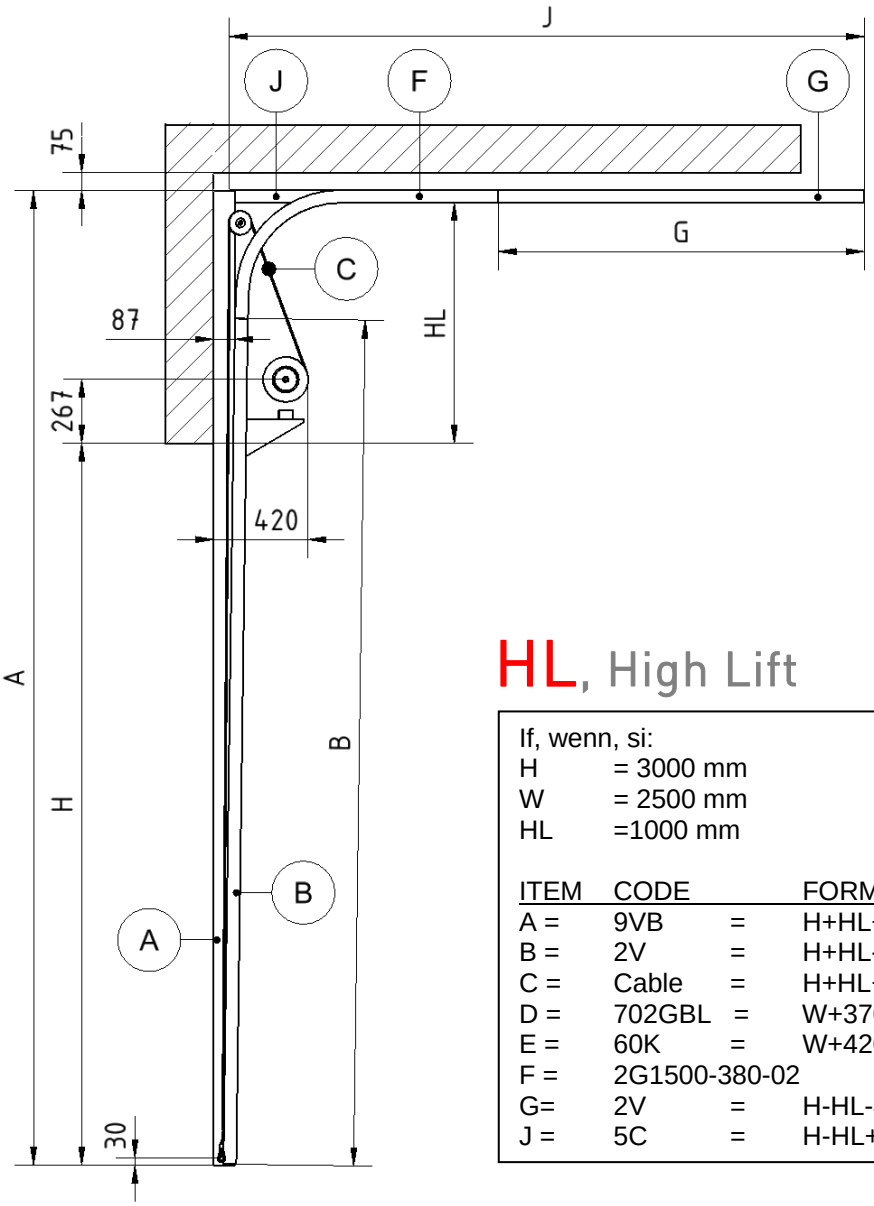


VL, Vertical Lift

If, wenn, si:
H = 3000 mm
W = 2500 mm

ITEM	CODE		FORMULA			
A =	9VB	=	(2xH)+295	=	6295	mm
B =	2V	=	(2xH)+130	=	6130	mm
C =	Cable	=	(2xH)+5000	=	11000	mm
D =	702GBL	=	W+370	=	2870	mm
E =	60K	=	W+420	=	2920	mm

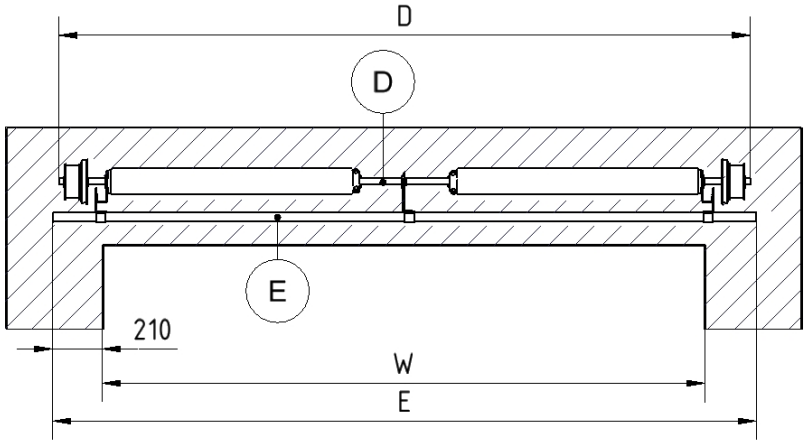




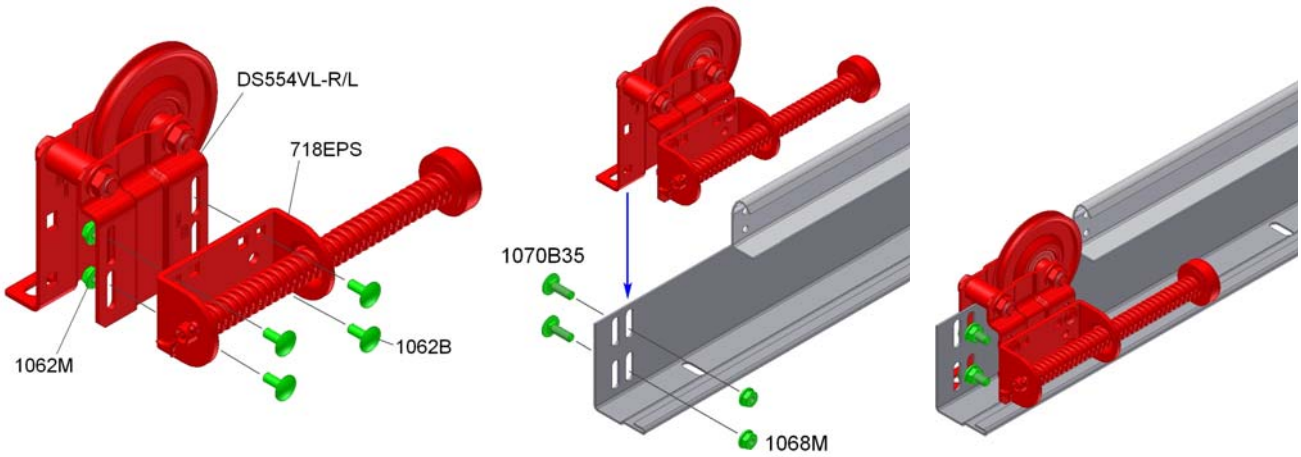
HL, High Lift

If, wenn, si:
H = 3000 mm
W = 2500 mm
HL =1000 mm

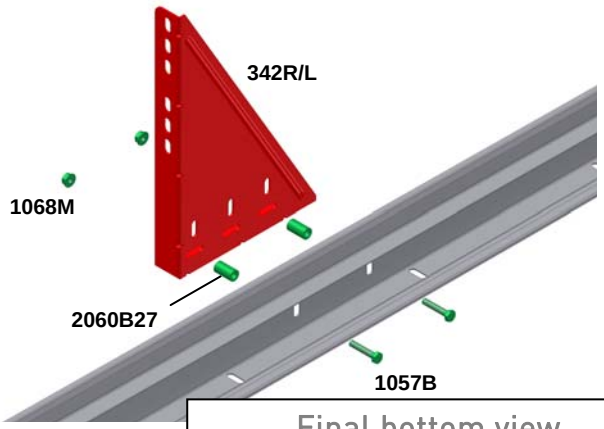
ITEM	CODE		FORMULA			
A =	9VB	=	H+HL+50	=	4050	mm
B =	2V	=	H+HL-485	=	3515	mm
C =	Cable	=	H+HL+2344	=	6344	mm
D =	702GBL	=	W+370	=	2870	mm
E =	60K	=	W+420	=	2920	mm
F =	2G1500-380-02					
G=	2V	=	H-HL-480	=	1520	mm
J =	5C	=	H-HL+695	=	2695	mm



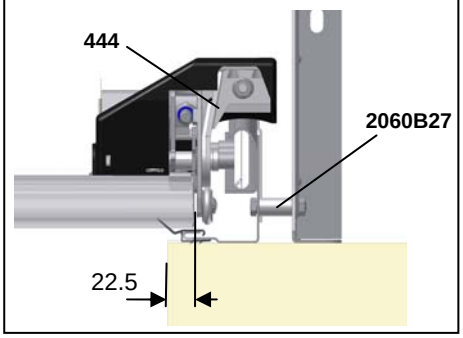
INSTALLATION VERTICAL LIFT RAIL SYSTEEM



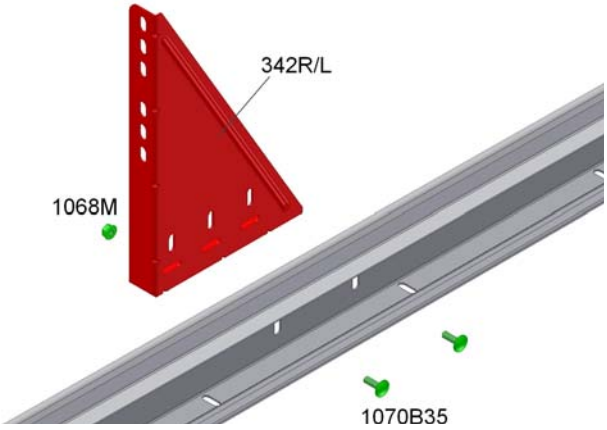
WHEN USING 444 CABLE BREAK:



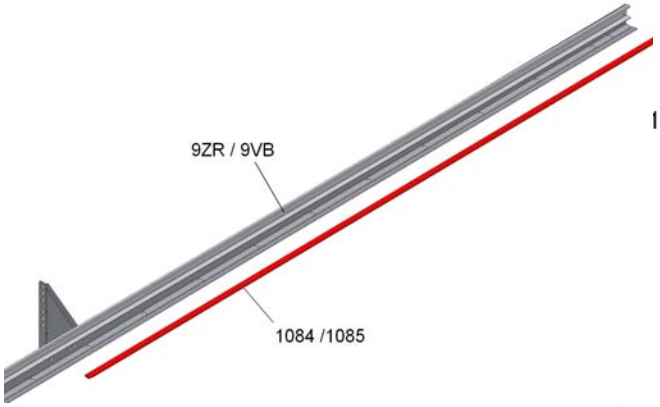
Final bottom view

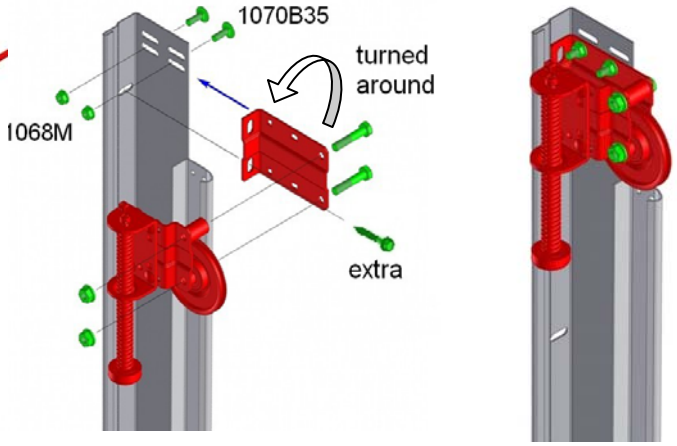


WHEN USING BOTTOM BRACKET:

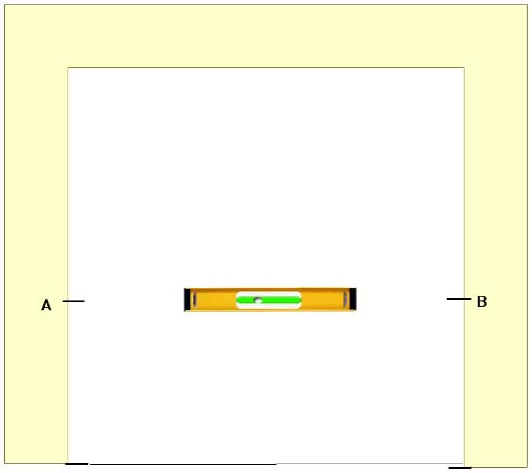


Alternative assembly DS554VL-R/L:

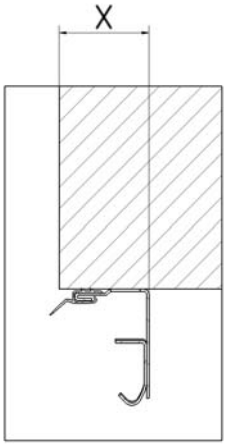


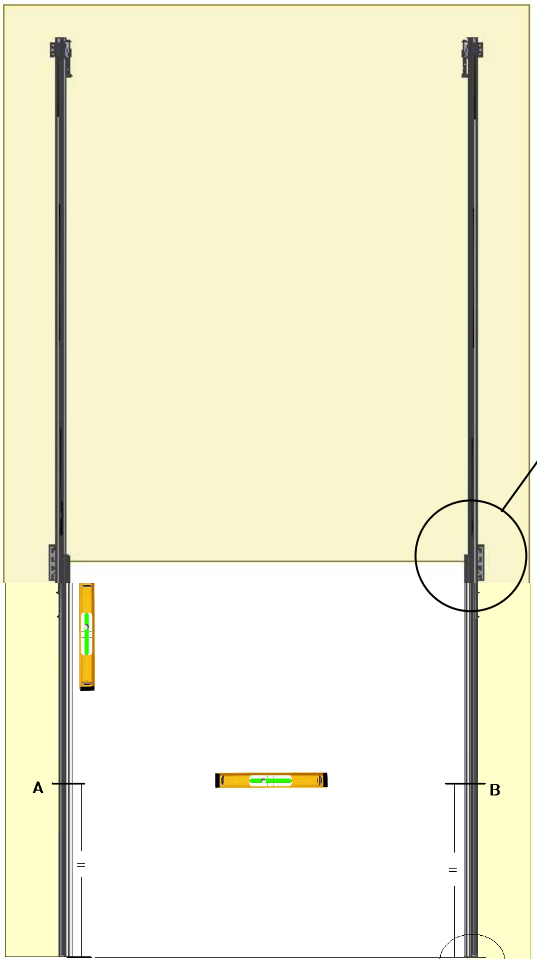


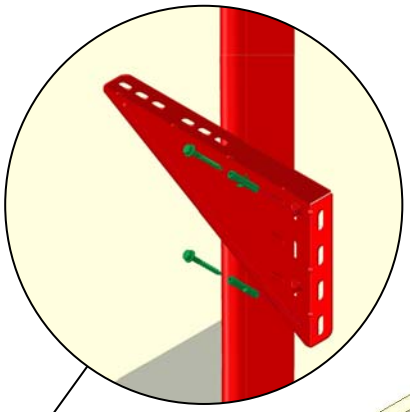
INSTALLATION VL, VERTICAL LIFT RAIL SYSTEEM

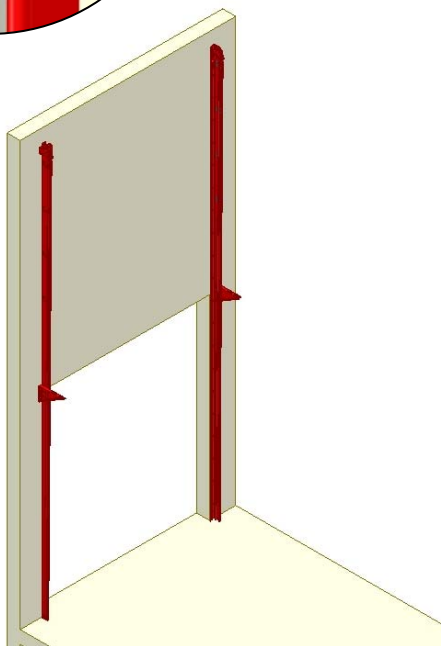


Type	X
427SX	70 mm
444	75 mm
429	65 mm



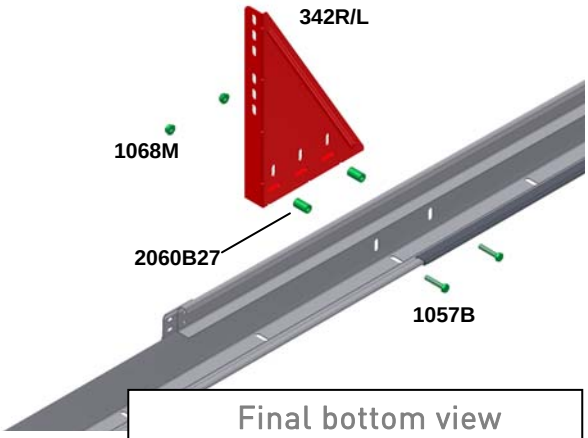


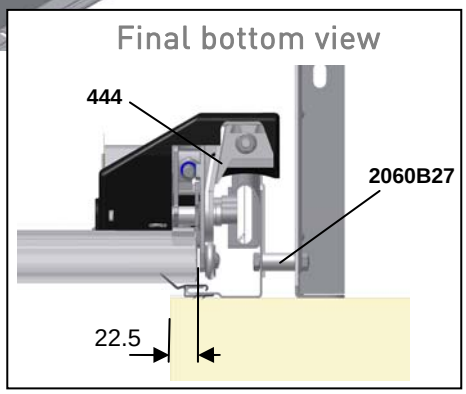




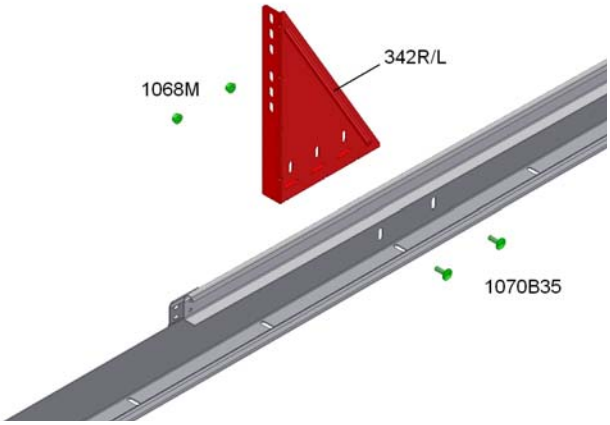
INSTALLATION VERTICAL LIFT RAIL SYSTEEM

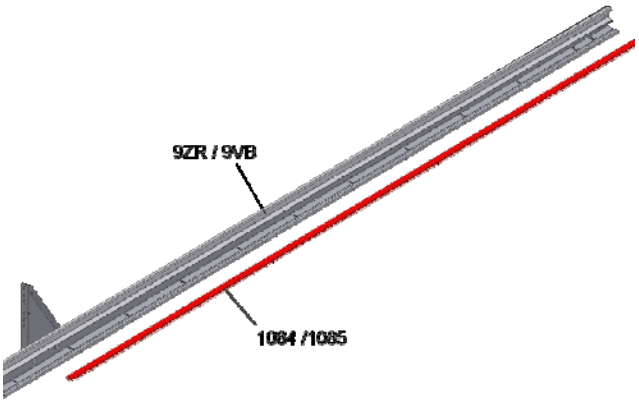
WHEN USING 444 CABLE BREAK:

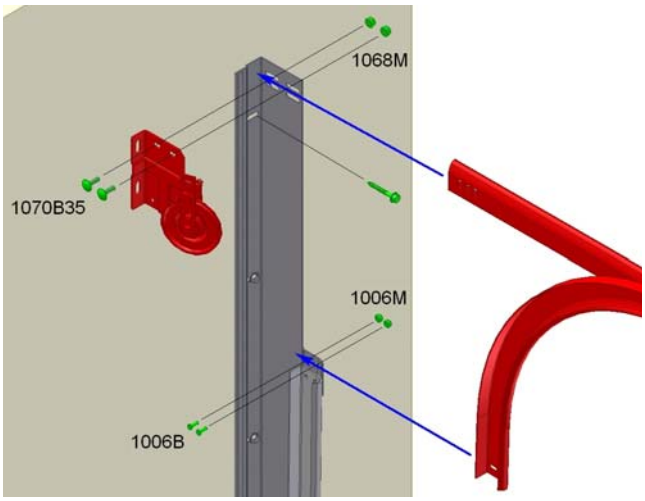


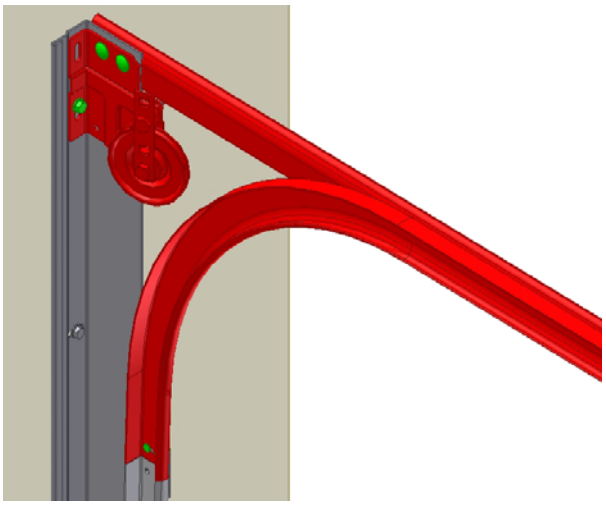


WHEN USING BOTTOM BRACKET:

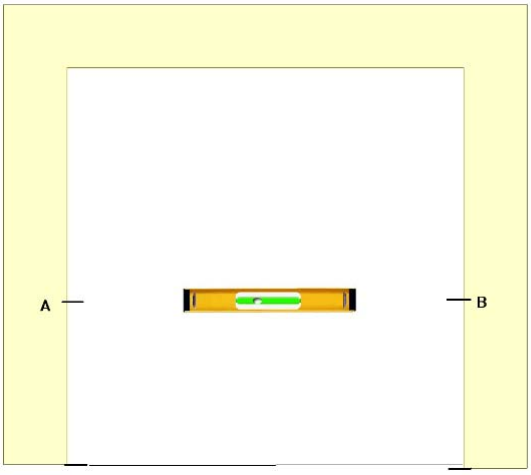




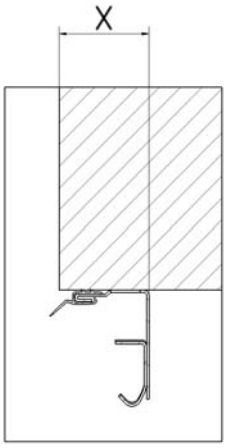


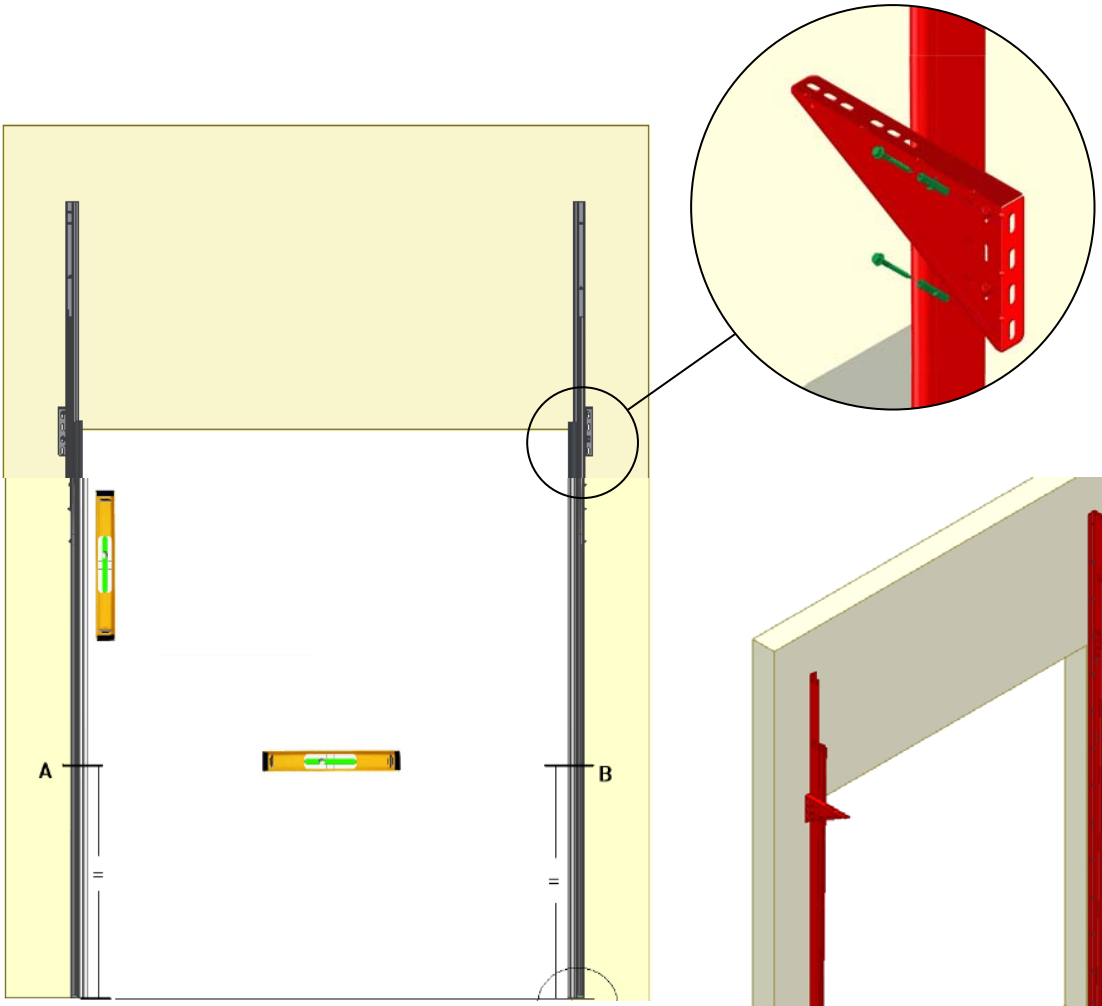


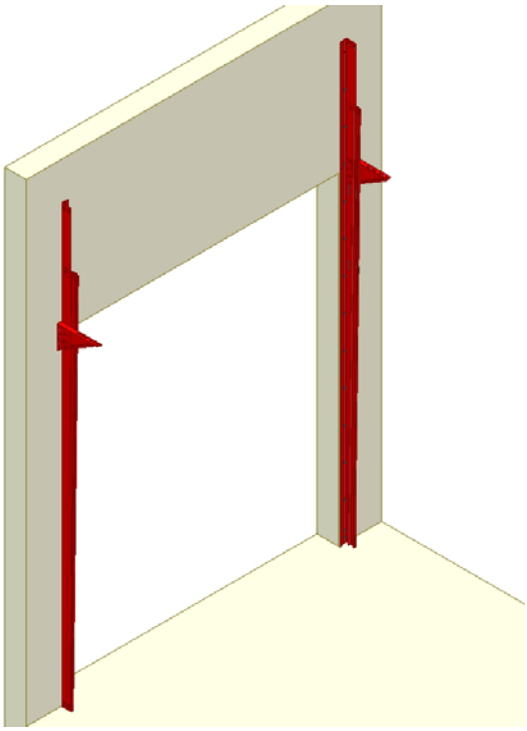
INSTALLATION HL, HIGH LIFT RAIL SYSTEEM



Type	X
427SX	70 mm
444	75 mm
429	65 mm

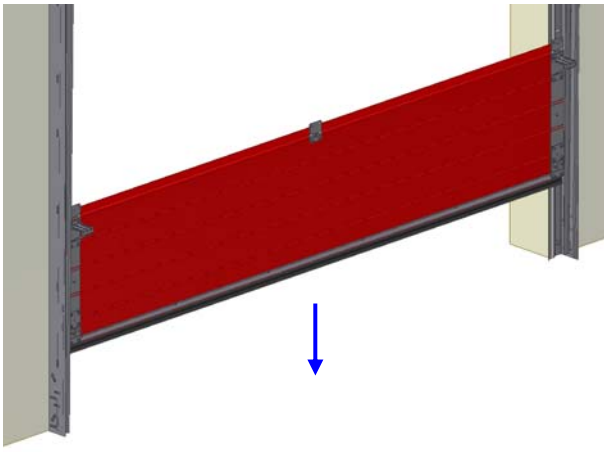


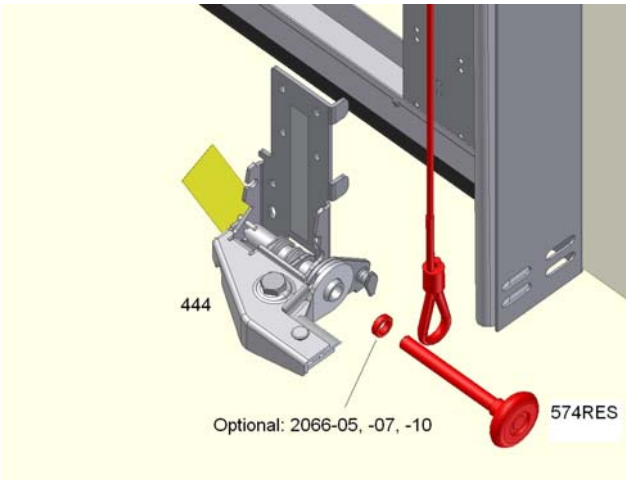


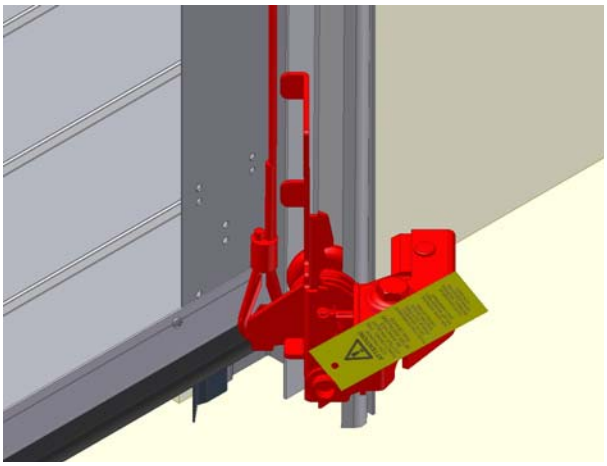


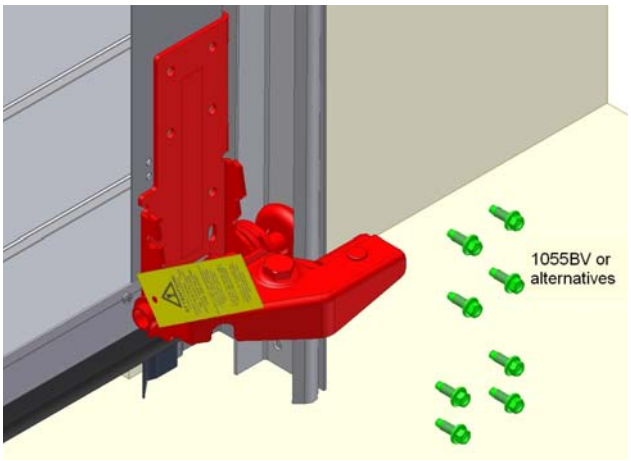
PANEL <> HARDWARE INSTALLATION

Assembly of panels see chapter Panel assembly.

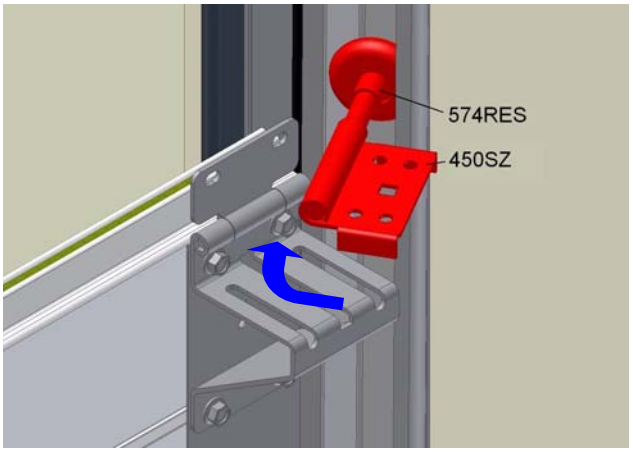


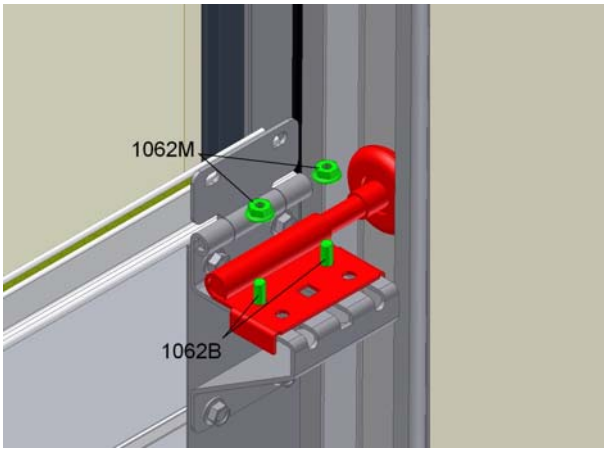


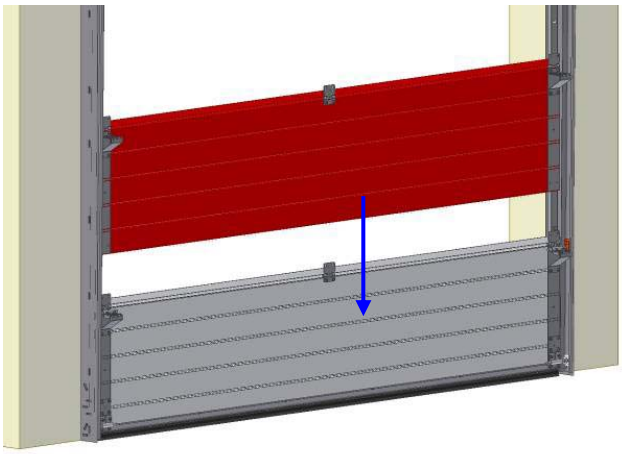


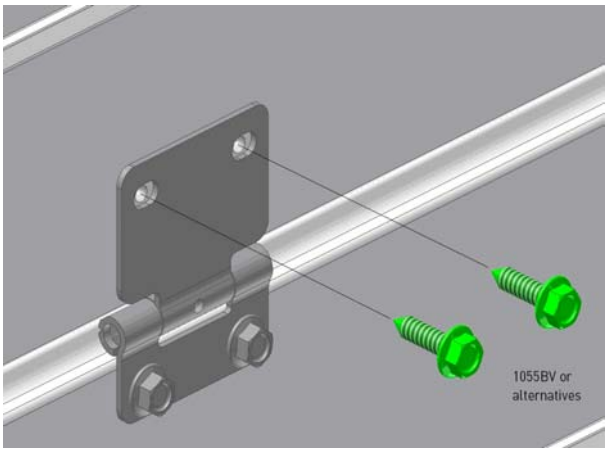


For detailed operation/installation of CBD see Manual 444.





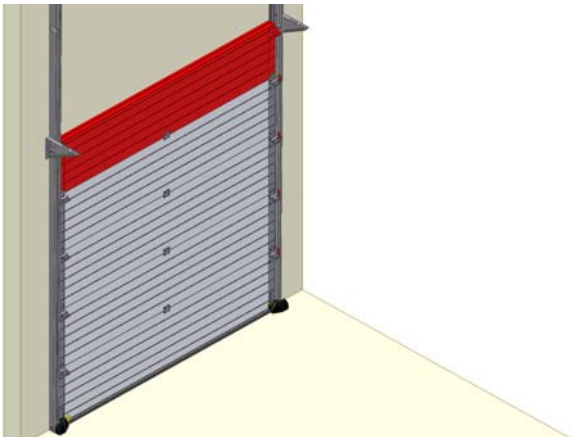


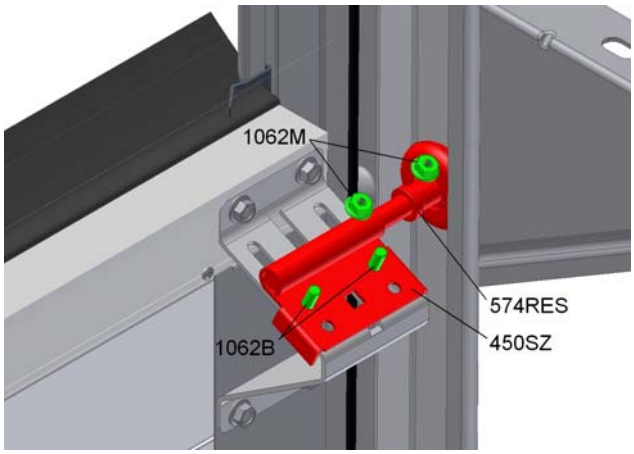


Attention! Cable runs to pulley behind the roller shafts.

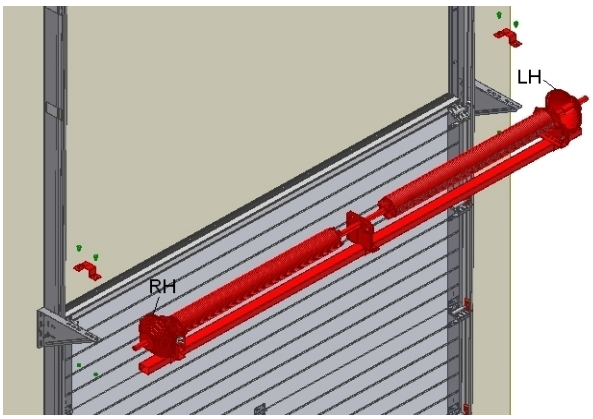
INSTALLATION TOP PANEL / TOP ROLLER CARRIER

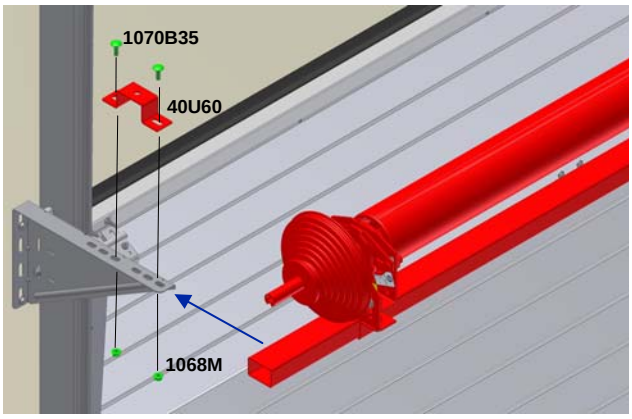
Close the door and secure the door panel. Press the top panel against the side (upper) seal. Turn the top roller holder into the track and slide the top roller holder as far as possible downwards and fix all screws (minimum play between door panel and seal).



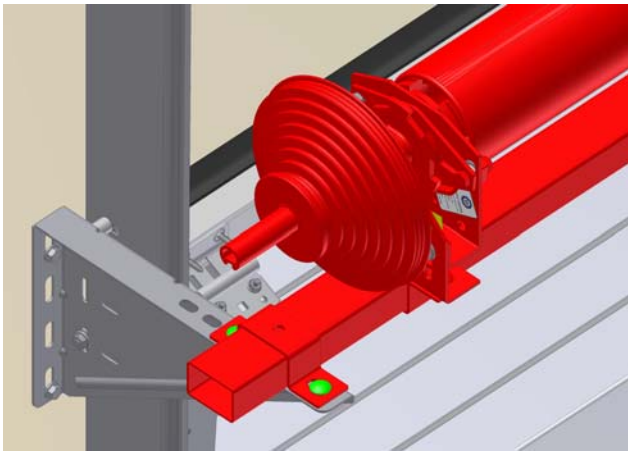


INSTALLATION SHAFT SYSTEM

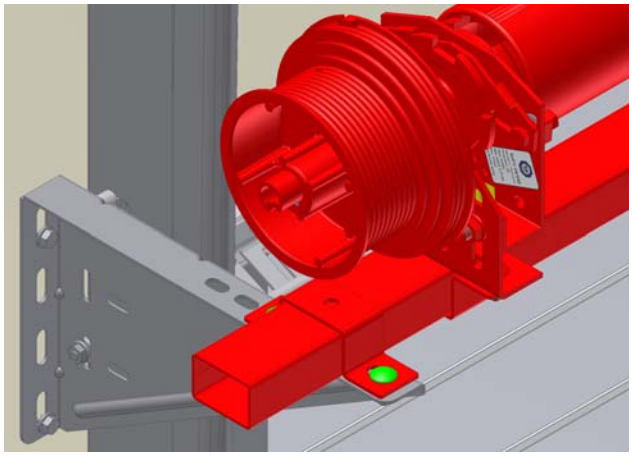




The Shaft System is pre-assembled. 2 torsion springs is standard. 1 torsion spring is optional.



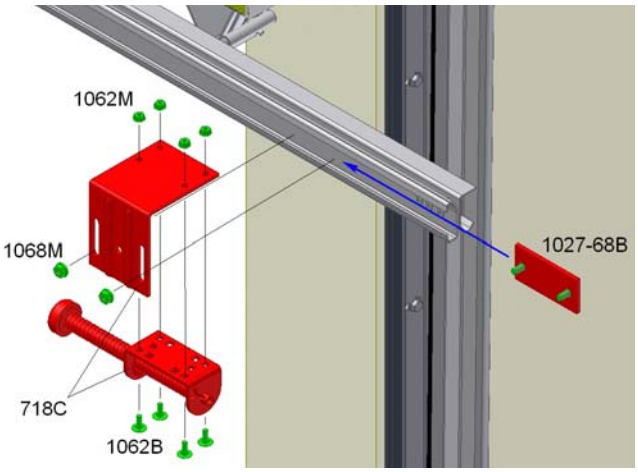
VL, Vertical Lift

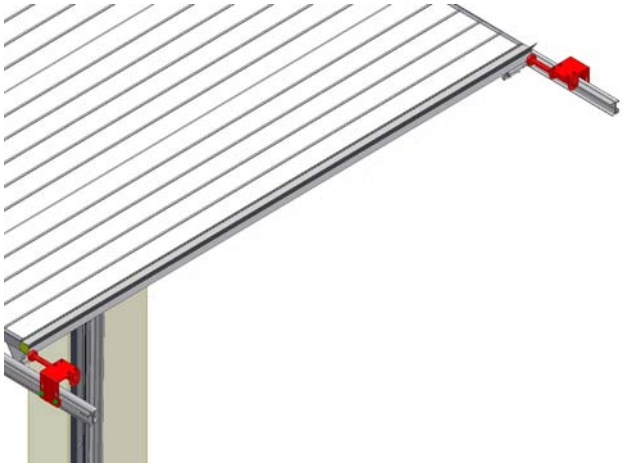


HL, High Lift

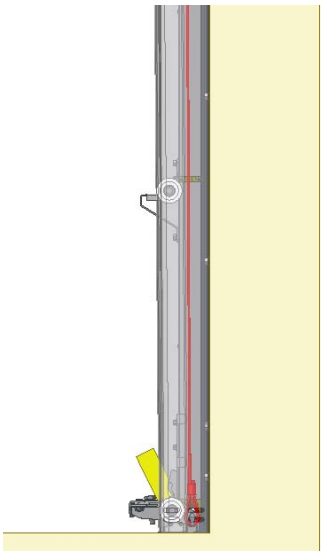
Spring break device as ordered: 670 or 675 or none.

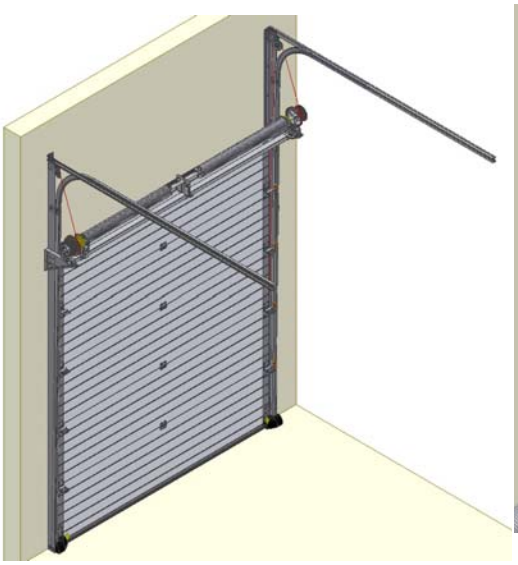
INSTALLATION SPRING BUMPER HIGH LIFT

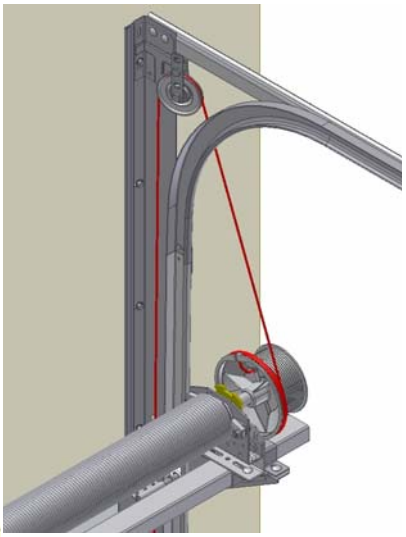




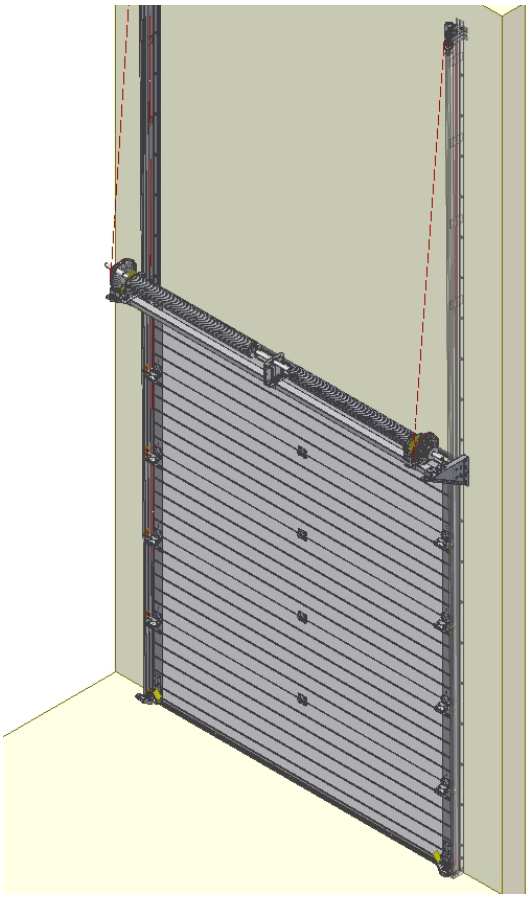
CABLES & TORSION SPRINGS

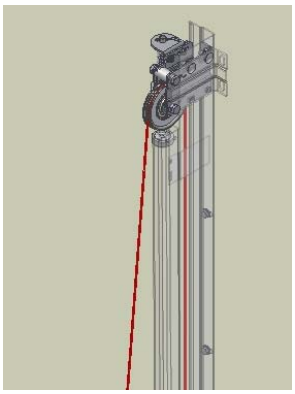


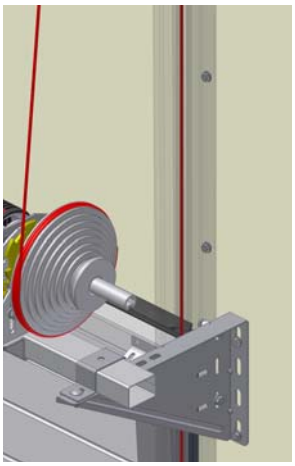




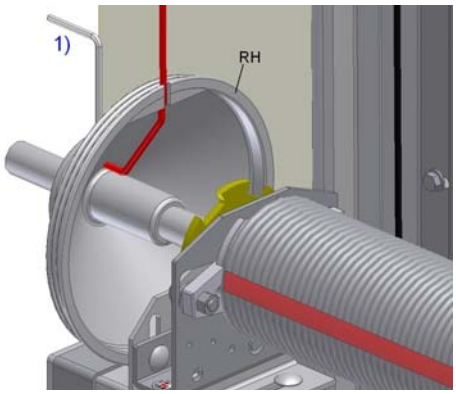
Attention! Cable runs to pulley behind the roller shafts.

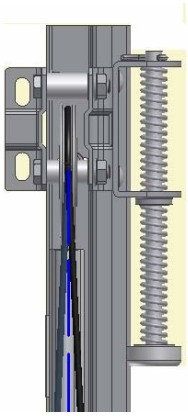


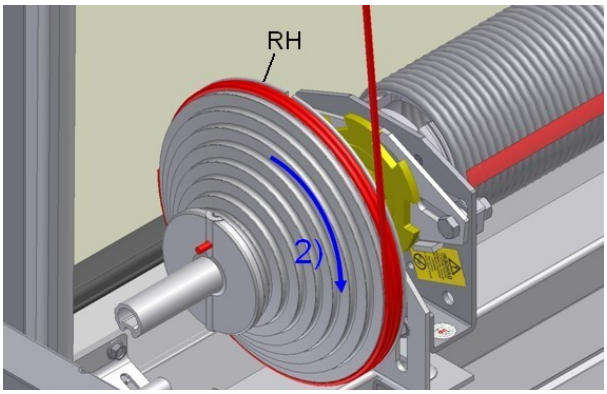


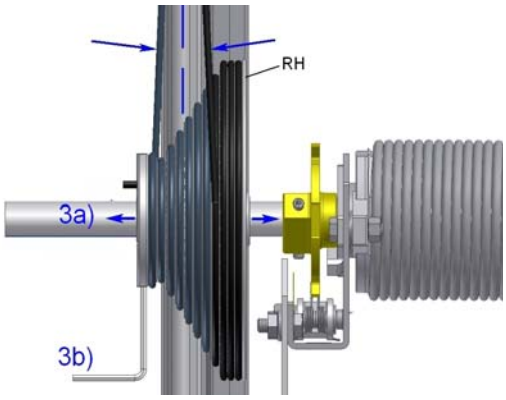


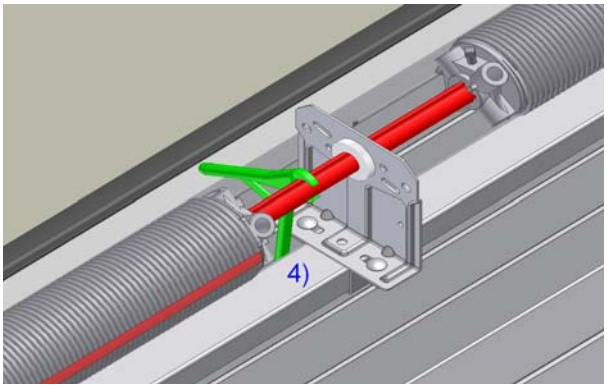
WINDING OF THE CABLE

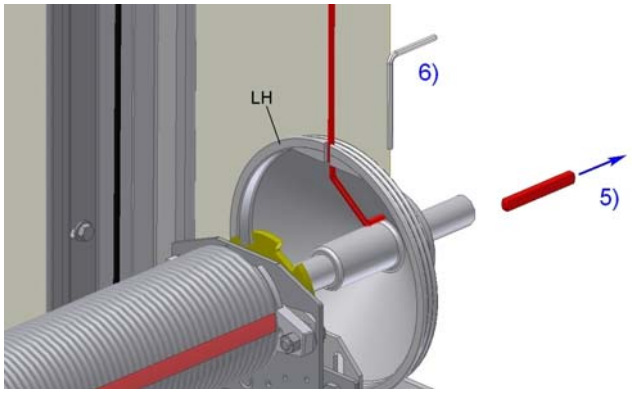


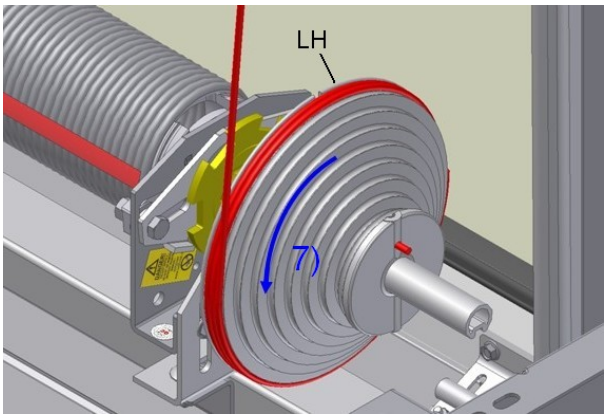


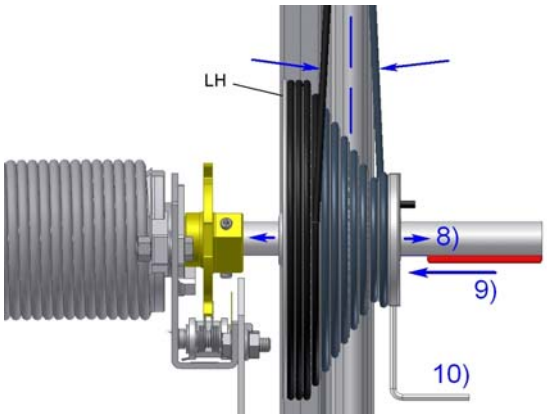










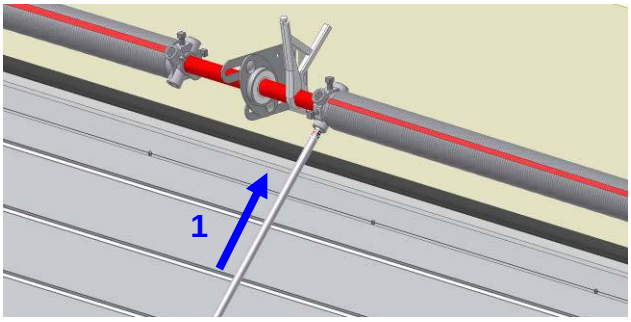


NB! See separate manual spring break device installation: 670Manual GB.pdf

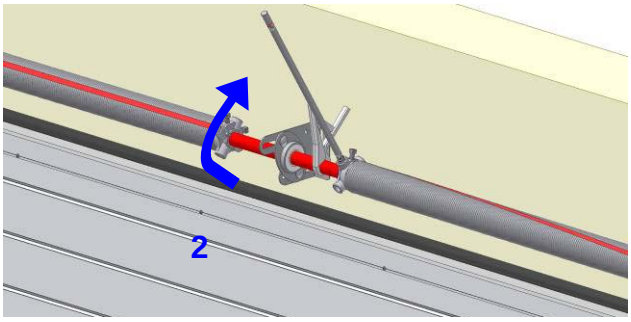
TORSION SPRING TENSIONING

CAUTION!

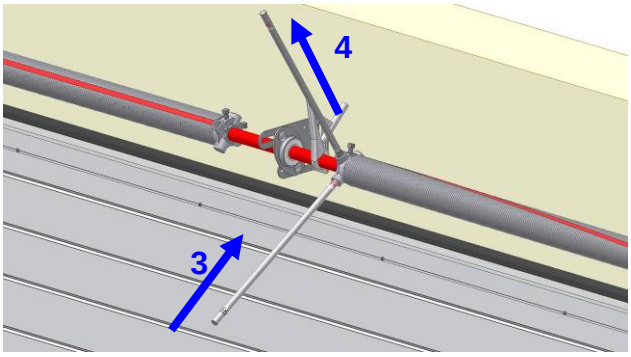
Torsion springs are subject to considerable tension! Proceed at all times with extreme caution. Installation, maintenance and repair should be carried out only by experienced and properly trained overhead door fitters. Use correctly fitting and properly maintained tension levers.



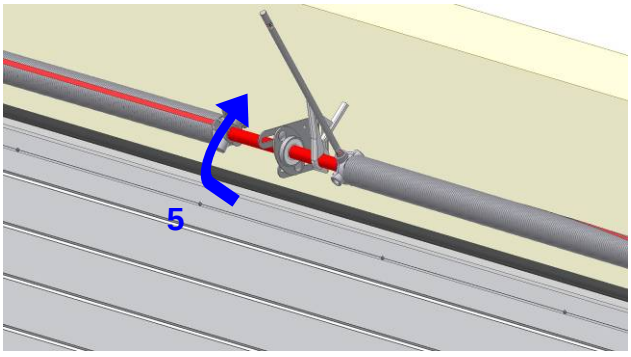
Insert the 1st tension lever completely into the slot.



Turn the 1st tension lever a quarter turn such that the spring is tensioned.

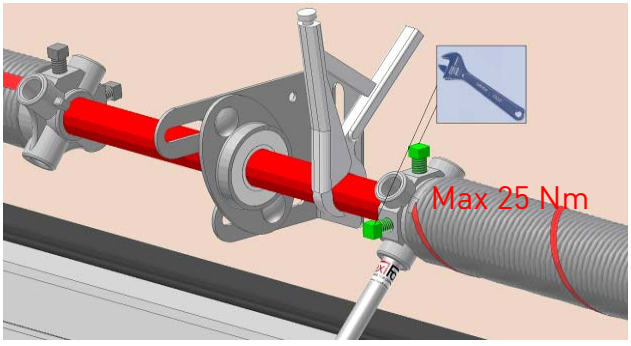


Insert the 2nd lever completely into the next slot and remove the 1st tension lever from the slot.

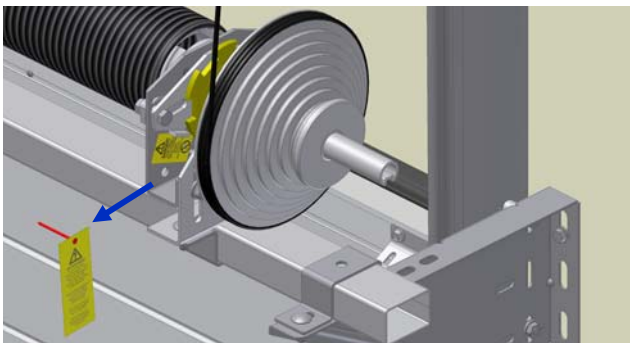


Turn the 2nd tension lever a quarter turn such that the spring is tensioned.

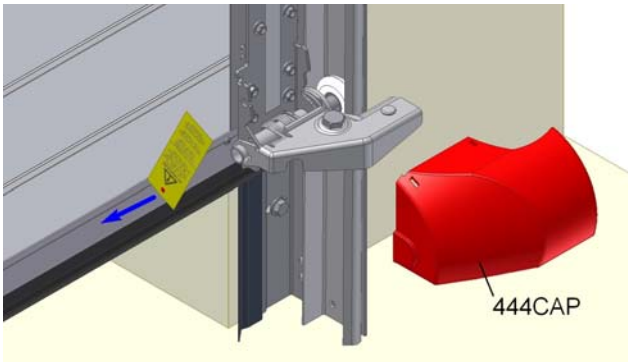
Repeat until number of turns is achieved.



Fix the spring plug to the shaft.



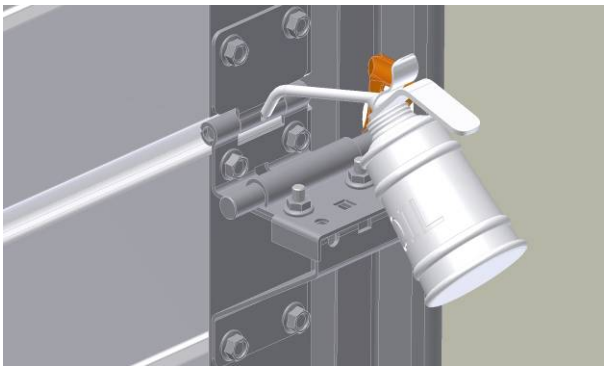
Remove the security pin from the spring break device (if fitted).

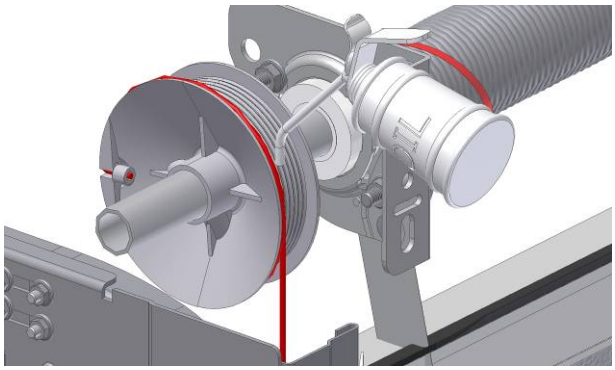


See manual 444: Remove the security pin form the cable break device (if fitted) and install the protective cover.

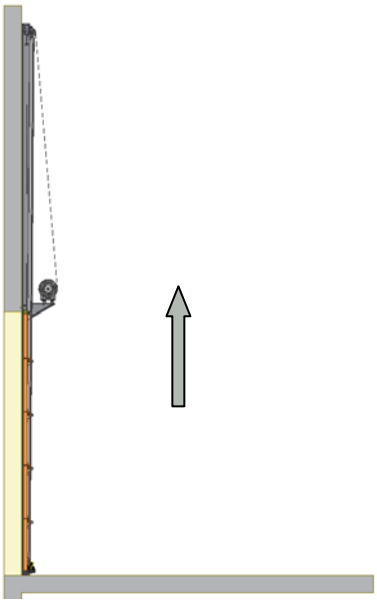
FINISHING THE DOOR

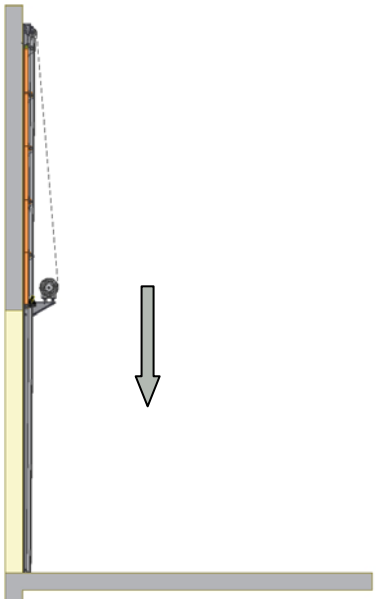
GREASING DOOR PARTS





RE-ADJUSTING THE DOOR

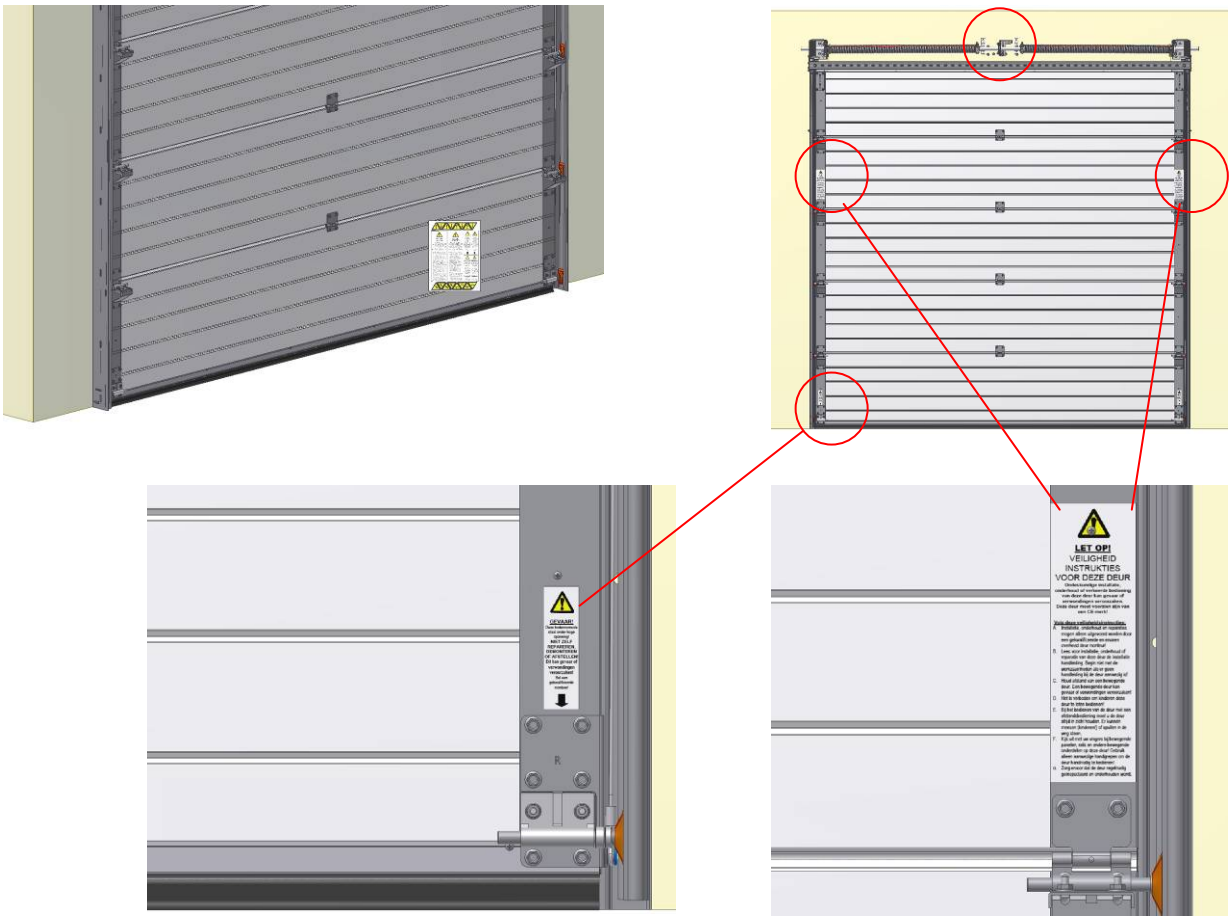




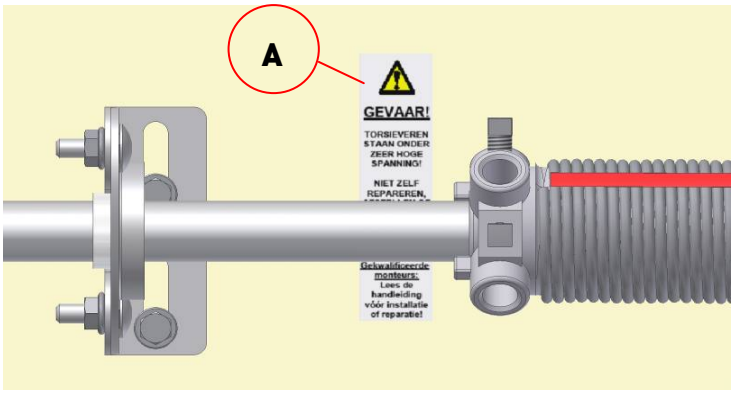
Open and close the door for checking all the settings.

When the door panel is not hanging completely horizontal in the lifting cables in (almost) closed conditions adjust the cable in the adjustable bottom bracket (if fitted) or adjust the cable in one of the drums.

POSITION SAFETY LABELS ON THE DOOR







Install any additional accessories that you have ordered separately such as:
Handgrip, lock, etc. See separate instructions in product packaging.

NB! A lock may not be fitted to an electrically driven door.

NB! A manually operated door should always be equipped with a hand grip, installed at a safe location on the door.

NB! DO NOT INSTALL A PULL ROPE WITH A POWER OPERATED DOOR AT ANY TIME!! THIS CAN CAUSE SERIOUS INJURIES OR DANGER TO THE USER OF THE DOOR!

Place your CE identification plate on the door (mandated!!) together with any warning labels required.

Hand over necessary documents to the end-user:

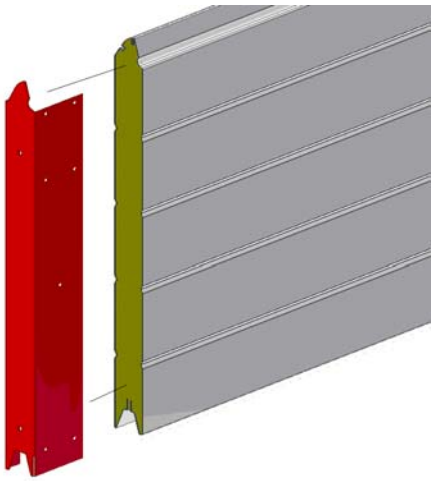
- User manual
- Dismantling instructions (included in this manual)
- Maintenance instructions (included in the user manual)
- Service log book
- Declaration of conformity IIa declaring the door is according to EN-13241-1.

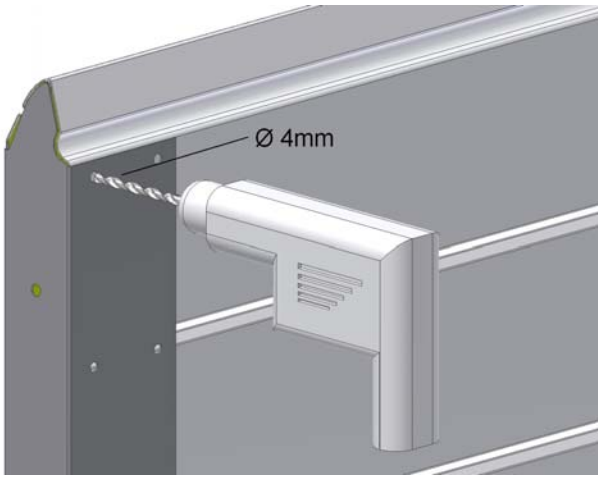
Visit www.flexiforce.com for the most recent update of information on your obligations towards handing over a safe-to-use door.

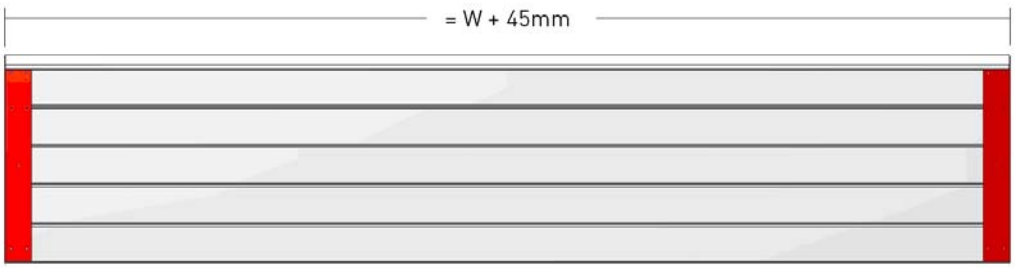
PANEL ASSEMBLY

INSTALLING END CAPS (RES- / IND-panels)

40E610 or alternatives

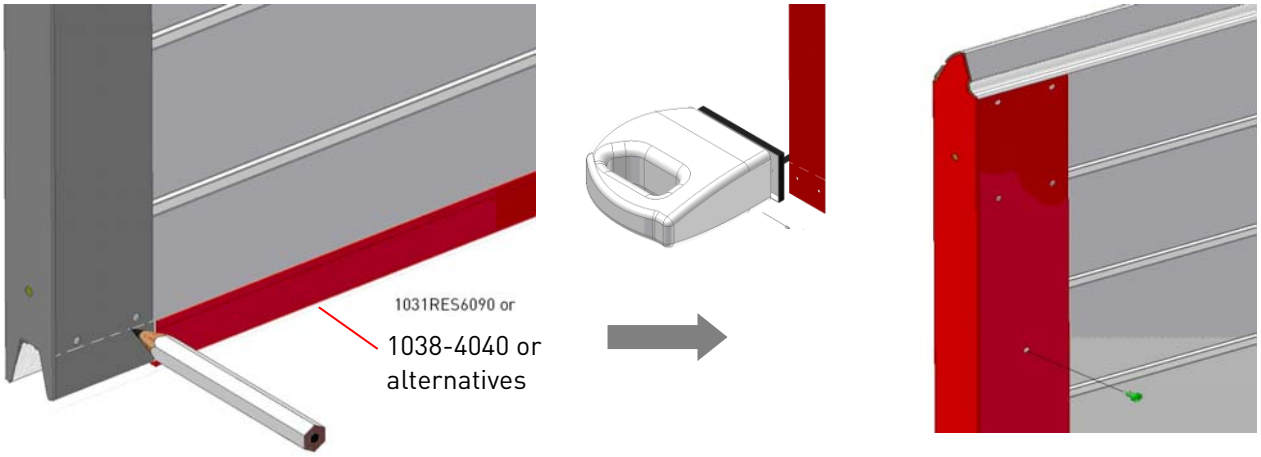


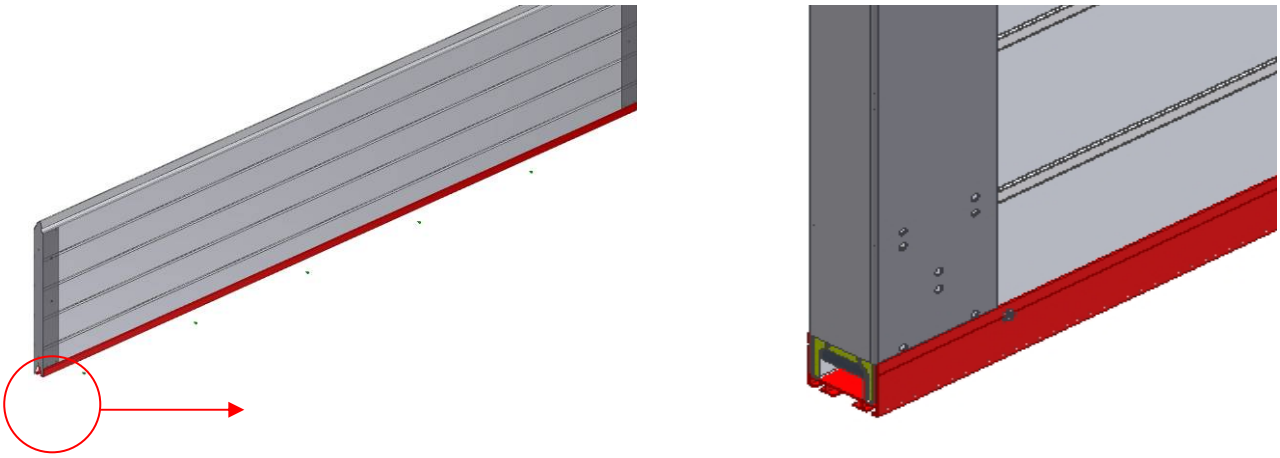


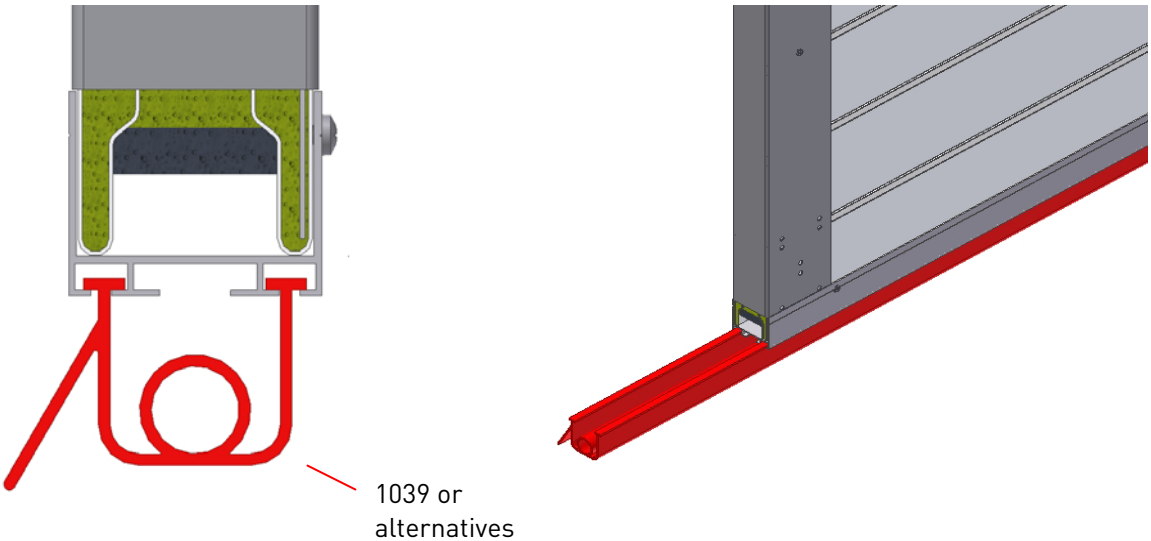


W = Days opening width

INSTALLING BOTTOM ALU + SEAL (RES- / IND-panels)



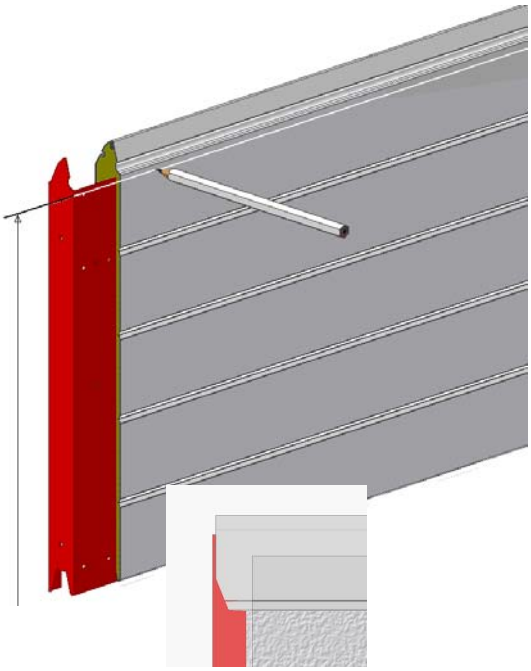


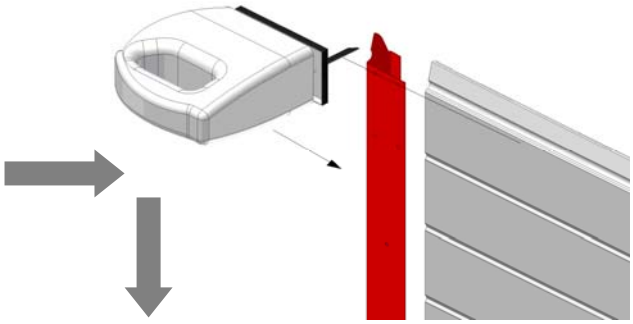


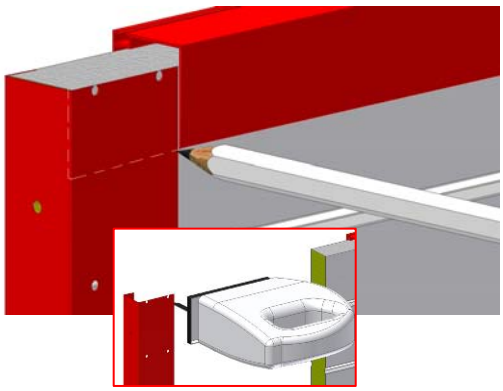
CUTTING / FINISHING TOP PANEL

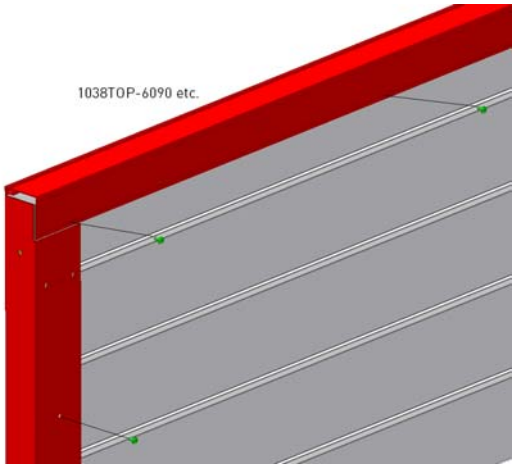
NB! Make sure that the bottom panel will not be cut for adjusting the panel or door height. This should be done with the top section only! The attachment of the bottom bracket, which is holding the complete door weight, is relying on the construction of the panel. Danger can occur if the panel is cut at this point.

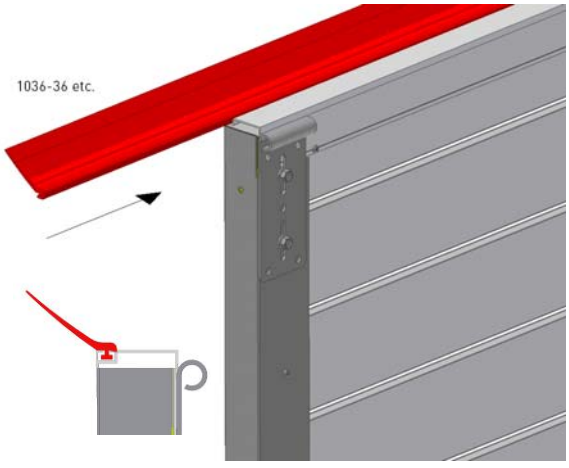
Total door panel height = Days Opening Height (H) + 25mm











TROUBLESHOOTING

What should be checked if the door is not balanced properly?

When a door is not well in balance, then it is necessary to check first the following details :

1. Is the given information correct :

   - weight of the door leaf (including hardware)
   - is the division of the weight equal on each panel, or are there panels with a different weight than the others, for instance by the application of different panels (glass, pass door with heavy profiles).

2. Are the correct parts supplied and fitted?

   especially the drums and springs are important :

   - are the correct dimensions supplied ?

3. Is the door properly installed?

   - horizontal tracks really horizontally and not with inclination.

4. Were modifications made afterwards ?

   - check if any changes were made during the fitting, or if a pass door was fitted later, or any reinforcement profiles fitted etc.

5. Is the e-operator installed and selected properly?

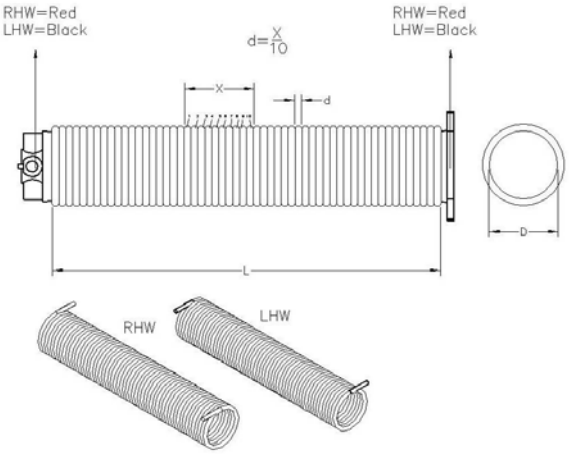
   - Is the installed e-operator suitable for this door?
   - Is the e-operator power adjusted properly for this door? (See installation manual operator)

WHAT TO DO AFTER CABLE BREAK OR SPRING BREAK?

NB! Also make sure to instruct the user about this.

After the braking of a spring or cable the door will be stopped by the appointed spring break or cable break device (if fitted). The end user must contact immediately a qualified overhead door mechanic. Spring break and cable break devices are a so called "one-shot" devices. After it has acted, it must be replaced, together will all other possibly damaged components of the door, such as torsion springs, spring plugs, bearing plates etc. The door has to be inspected thouroughly. See separate manuals of the fitted spring break and cable break device.

IDENTIFICATION SPRINGS







FlexiForce® spring gauge: WGD10

DISMANTLING THE OVERHEAD DOOR

ATTENTION! WARNING!

To dismantle an existing overhead door, a number of precautions must be taken. For the safety of all concerned pay heed to the warnings and instructions given below! If in doubt, contact your supplier.
Dismantling should only be carried out by experienced fitters. This manual is not suitable for d.i.y. purposes or for use by trainee fitters.
This manual only describes the installation/dismantling of hardware for overhead doors and as such must be supplemented with instructions for any additional components.
FOR ANY DETAILS ON THESE DISMANTLING INSTRUCTIONS, WE REFER TO THE INSTALLATION CHAPTERS OF THIS MANUAL WHERE DRAWINGS AND DETAILS ARE DISPLAYED.

STEP 1. De-tensioning the torsion spring(s)

CAUTION ! Torsion springs and bottom brackets are under high tension. Exercise at all times great caution. Use properly fitting and maintained tension irons (see drawing).

Start dismantling of the door by closing the door and securing its movement with a clamp on the vertical track. First the tension on the torsions springs and cable has to be released. Do this by following these instructions :

1 Insert the 1st tensioning iron fully into the tensioning aperture.
2 Take over the tension of the spring with this tensioning iron.
3 Loosen the bolts in the tensioning plug and remove the key.
4 Turn the 1st tensioning iron in the direction required.
5 Insert the 2nd tensioning iron fully into the next tensioning aperture.
6 Take over the tensioning of the spring from the 1st tensioning iron with the 2nd tensioning iron.
7 Remove the 1st tensioning iron from the aperture.
8 Turn the 2nd tensioning iron a quarter turn in the direction required.
9 Insert the 1st tensioning iron fully into the tensioning aperture.
10 Take over the tensioning of the spring from the 2nd tensioning iron with the1st tensioning iron.
11 Repeat steps 4 through 10 until all tension is released.
12 Remove the last tensioning iron.

STEP 2. Disconnect the electrical operator. Follow any instructions given in the separate manual of the operator.
STEP 3. Loosen the cable drums and remove the keys. Act carefully, there might be some tension left on the cable. Check if the cable is slack. Remove the cable by disconnecting it from the bottom bracket and cable drum.
STEP 4. Dismantle the horizontal track construction.
STEP 5. Remove the panels one by one from the vertical track construction, starting with the top panel. Do this by loosening the hinges and rollers first.
STEP 6. Remove the shaft construction, after you have dismantled the E-operator from the shaft. If the shaft is divided and connected with a coupler, first disconnect the coupler and carefully remove both halves of the shaft system.
Attention ! Watch out for parts that might slide of the shaft, such as cable drums, bearings or keys.
STEP 7. Remove vertical tracks and angles from the building construction.
STEP 8. Make sure that you remove all the parts and panels in an environment kindly way. Check with your local authorities where and how you can leave this as garbage.

MAINTENANCE AND REPLACEMENT OF PARTS DOCK DOORS

An overhead door should be maintained and checked regularly to ensure safe operation and use. This is described in the EN-norms.

GENERAL:

1 Torsion springs, brackets and other components which are attached to the springs and cables, are under extreem tension. If not handled properly, injuries or damages might occur!
So, working on these components may only be carried out by qualified overhead door mechanics!

2 Replacement of broken or weared components should always be done by qualified overhead door mechanics.

3 When checking the door, always disconnect the electrical main power supply. Make sure that it is blocked against re-engaging without you knowing it.

REGULAR MAINTENANCE:

After installation:

1. Grease running part of the tracks	MECHANIC
2. Grease the bearings of the rollers	MECHANIC
3. Grease the shafts of the rollers	MECHANIC
4. Grease the bearings of the shaft	MECHANIC
5. Grease the hinge pins	MECHANIC
6. Grease the lock	MECHANIC
7. Protect the panels with carwax	USER
8. Grease the rubbers slightly with vasaline	USER

After 3 months:

1. Complete inspection visualy	MECHANIC
2. Check balancing system and adjust if needed	MECHANIC
3. Grease all the above mentioned points if needed	MECHANIC

Every 6 months (or after every 750 cycles):

1. Check side seals on damage or wear and tear	USER
2. Check top seal on damage or wear and tear	USER
3. Check bottom seal on damage or wear and tear	USER
4. Grease all above mentioned points	USER
5. Clean the panels	USER
6. Clean the windows (only water wash, do not use cloth)	USER
7. Remove dirt and waste from the door or its surroundings	USER

Every 12 months (or after every 1500 cycles):

1. Check or test the fixation of the springs to the fittings	MECHANIC
2. Check the balance of the door and adjust if needed	MECHANIC
3. Check the cables for damage or wear and tear	MECHANIC
4. Check the cable connection points on drums and bottom bracket	MECHANIC
5. Check the roller on wear and free moving space	MECHANIC
6. Check the hinges on breaking	MECHANIC
7. Check the panels on damage, wear and roust	MECHANIC

8. Check the spring break device acc. to instructions in manual	MECHANIC
9. Check the cable break device acc. to instructions in manual	MECHANIC
10. Check and test the safety edge system with operator	MECHANIC
11. Check the manual operation of the door	MECHANIC
12. Grease the springs	MECHANIC

After two years (or after every 3000 cycles):

1. Grease all the above mentioned points	MECHANIC
2. Check or test the fixation of the springs to the fittings	MECHANIC
3. Check the balance of the door and adjust if needed	MECHANIC
4. Check the cables for damage or wear and tear	MECHANIC
5. Check the cable break device acc. to instructions in manual	MECHANIC
6. Check the cable connection points on drums and bottom bracket	MECHANIC
7. Check the roller on wear and free moving space	MECHANIC
8. Check the hinges on breaking	MECHANIC
9. Check the panels on damage, wear and roust	MECHANIC
10. Check the spring breaking device acc. to instructions in manual	MECHANIC
11. Check and test the safety edge system with operator	MECHANIC
12. Check the manual operation of the door	MECHANIC
13. Grease the springs	MECHANIC
14. Check side seals on damage or wear and tear	MECHANIC
15. Check top seal on damage or wear and tear	MECHANIC
16. Check bottom seal on damage or wear and tear	MECHANIC
17. Check the shaft on wear and tear or damage	MECHANIC
18. Check the bottom bracket on wear and tear and damage	MECHANIC
19. Check the connection of the drum to the shaft (keys!)	MECHANIC
20. Check and re-fix the bolt of the coupler	MECHANIC
21. Check the connections of the track system	MECHANIC
22. Check the suspension of the door to the lintel and ceiling	MECHANIC

Use for greasing : PTFE or SAE20 or WD40
Use for cleaning : Soft soap with water. Do not use aggressive soap or cloth.